

Bangladesh University of Engineering and Technology

Department of Chemistry

CHEM113: Chemistry

For Students of CSE

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CHEM113 : Chemistry

Total Credit Hours: 3 ; L1/T2

Quantum concept in atomic structure, VSEPR; molecular geometry, Quantum concept in bonding; VBT and MOT, Frontier MOT and electronic transition, Silicon chemistry, Properties of solutions, Colloid and Nano-chemistry, ~~Phase rule and phase diagram; Energy and chemistry, Electrochemistry; electrolytic conduction,~~ corrosion, devices for energy storage, Chemistry of biodegradable and conductive polymer; LED, LCD/touch screen, Chemistry of proteins, nucleic acids (DNA, RNA), carbohydrates and lipids; Introduction to computational chemistry; Design of new molecules, materials and drug.

Content

Corrosion and its consequences

Theory of corrosion

Types of corrosion

Crevice, Pitting, Waterline

Factors affecting corrosion

Corrosion prevention

Metal coating, Anodic protection

Cathodic protection, Electroplating



Corrosion and its consequences

Corrosion can be defined as the deterioration of a material's properties due to its interaction with the surrounding environment, including air, water, and chemicals.

Characteristics

- ❑ A naturally occurring process
- ❑ Materials degrade due to chemical reactions with their environment
- ❑ Weakening of materials and structures

Consequences of corrosion

Corrosion can lead to-

- ✓ structural failure
- ✓ reduced material lifespan
- ✓ safety hazards
- ✓ significant economic losses.

Theory of Corrosion

The overall theory is known as electrochemical theory of corrosion. It is also known as wet theory of corrosion because corrosion takes place in presence of moisture.

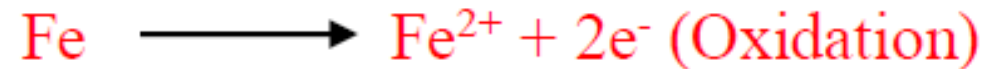
When metals are exposed to air, moisture, soil etc. then they undergoes corrosion through electrochemical process. In this process a separate anode and cathode areas are formed on the metal surface. Corrosion always occurs at the anodic area of metal through oxidation reaction and release electron. The liberated electron from anode, move towards cathode, where reduction take place. The flow of free electrons in this corrosion takes place through following processes:

- i) Hydrogen liberation
- ii) Oxygen absorption

Theory of Corrosion: *Hydrogen liberation*

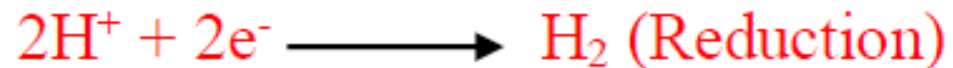
This type of corrosion evolution of hydrogen occurs in acidic medium. Let's considering the metal Fe, anodic reaction is dissolution of iron as ferrous ions with Liberation of electrons.

➡ At Anode

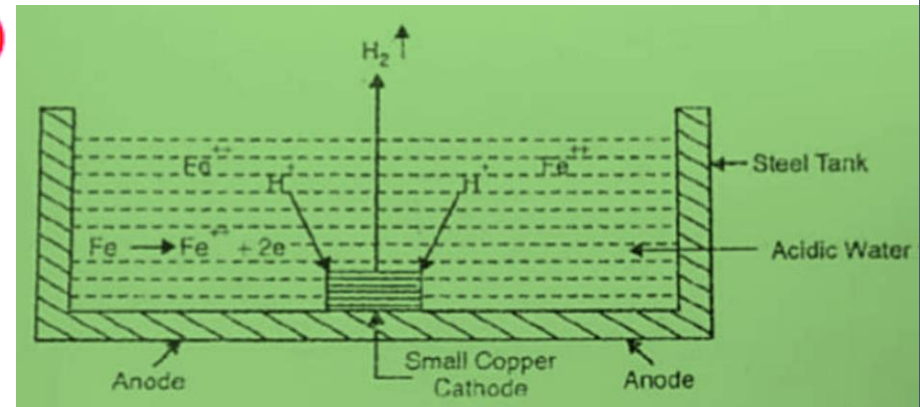


The electrons released flow through the metal from anode to cathode, whereas H^+ ions of acidic solution are eliminated as hydrogen gas.

➡ At Cathode



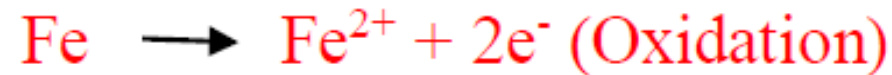
The overall reaction is:



Theory of Corrosion: *Oxygen absorption*

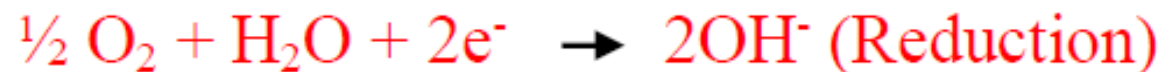
The rusting of iron takes place in neutral aqueous solution of electrolytes (presence of atmospheric oxygen). If the metal surface develops cracks, anodic areas are created on the surface whereas the metal parts act as cathodes.

➡ At anode

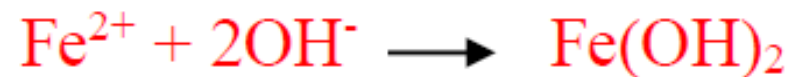


The released electrons flow from anode to cathode through iron metal.

➡ At cathode



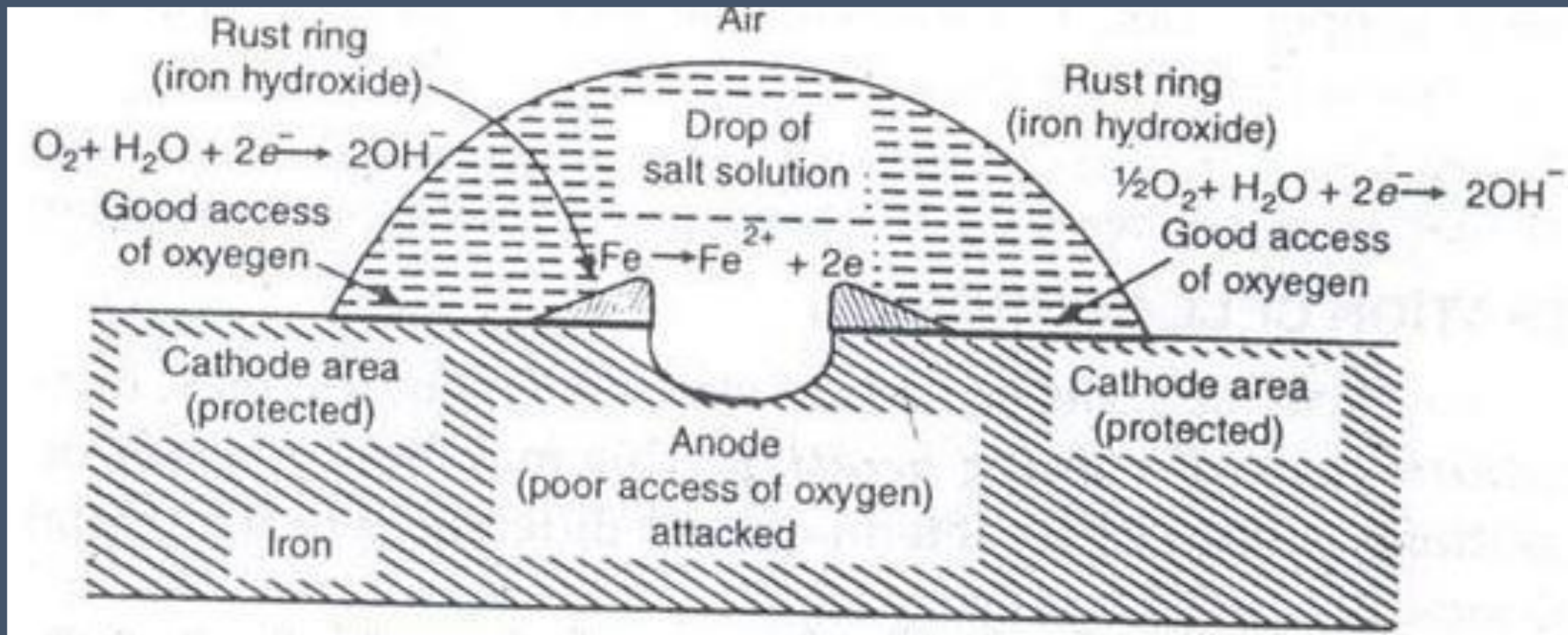
$\text{Fe}(\text{OH})_2$ precipitate when Fe^{2+} and OH^- ions reacts with each other



If oxygen is in excess, ferrous hydroxide is easily oxidized to ferric hydroxide.

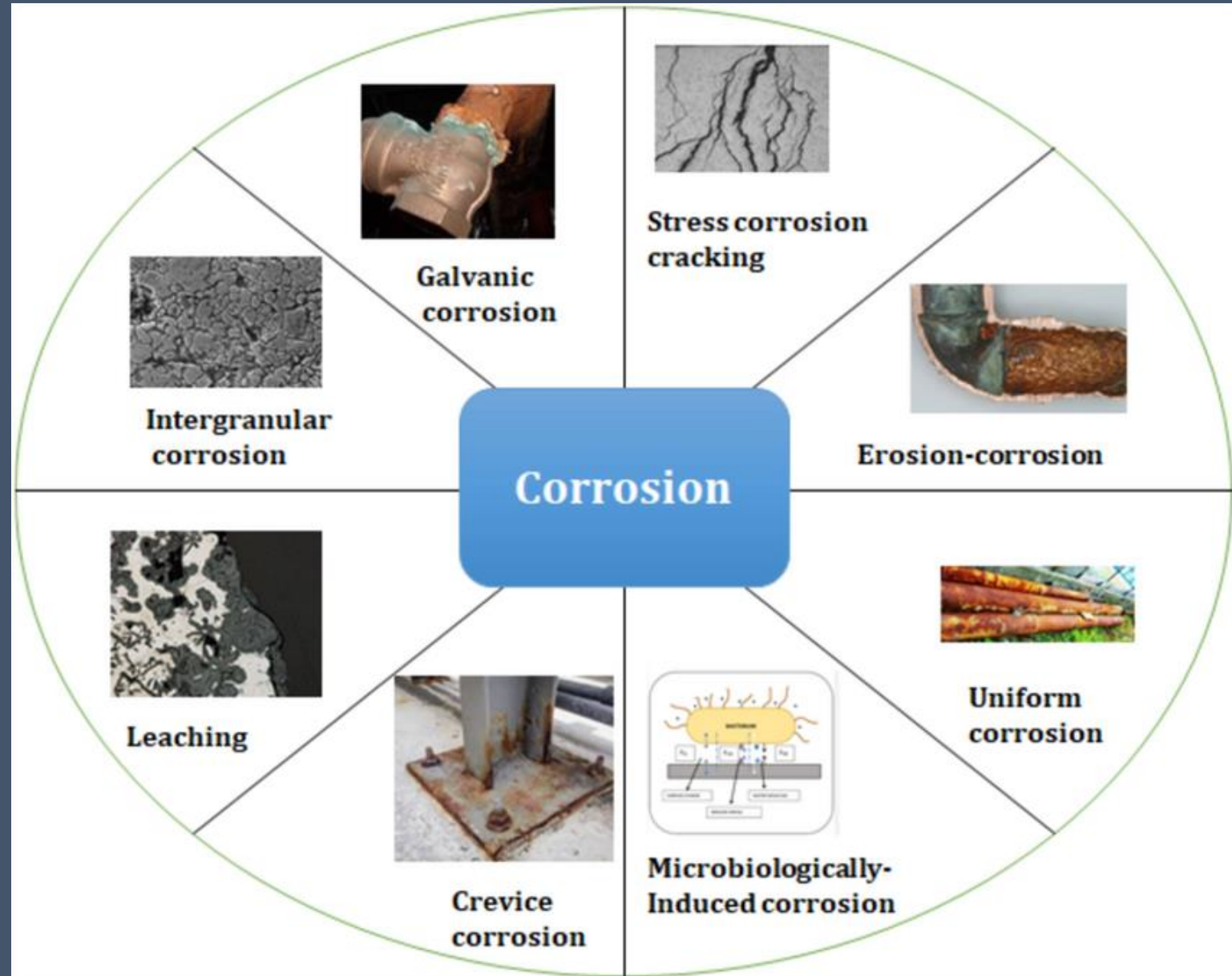


Theory of Corrosion: *Oxygen absorption*



Types of Corrosion

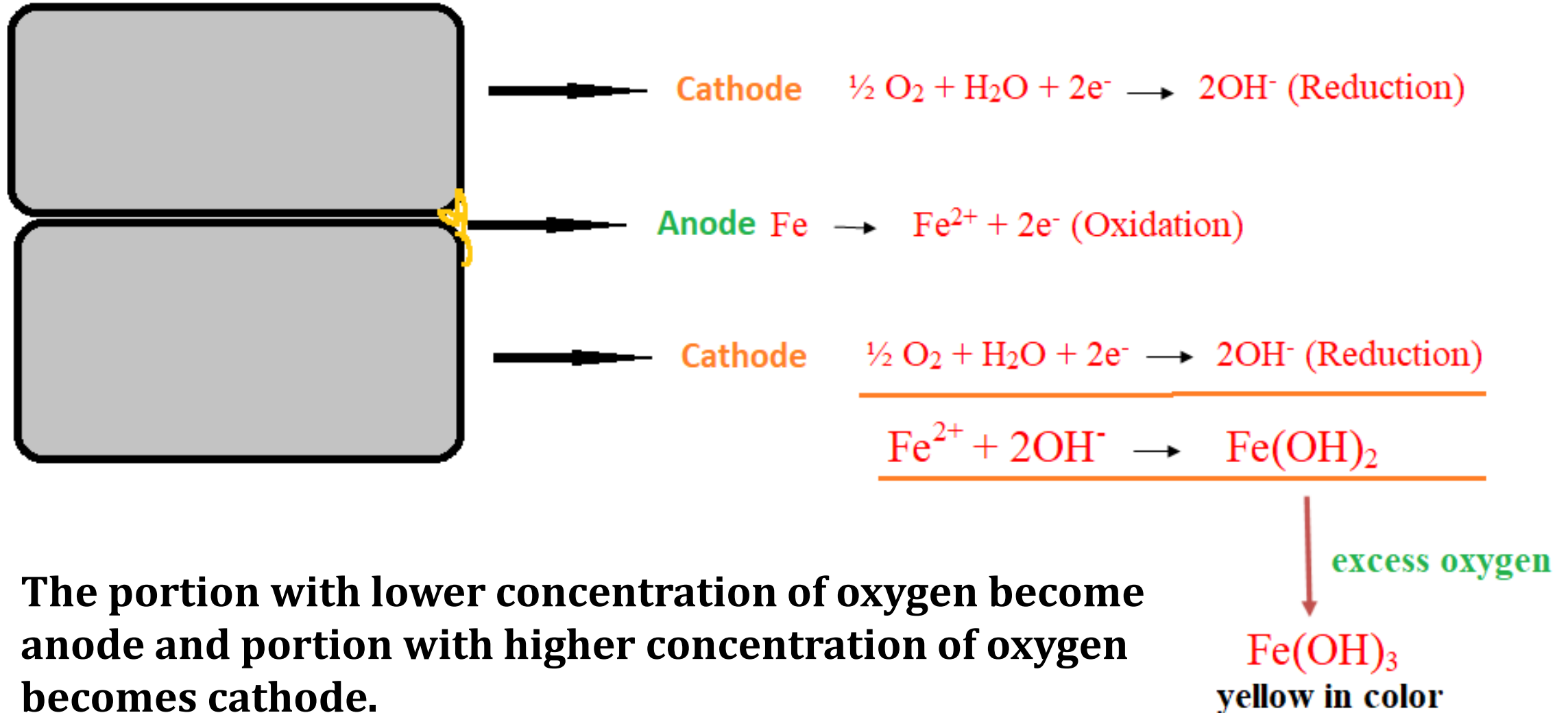
Corrosion can manifest in various forms, including uniform corrosion (affecting the entire surface), localized corrosion (like pitting or crevice corrosion), and corrosion fatigue.



Crevice corrosion

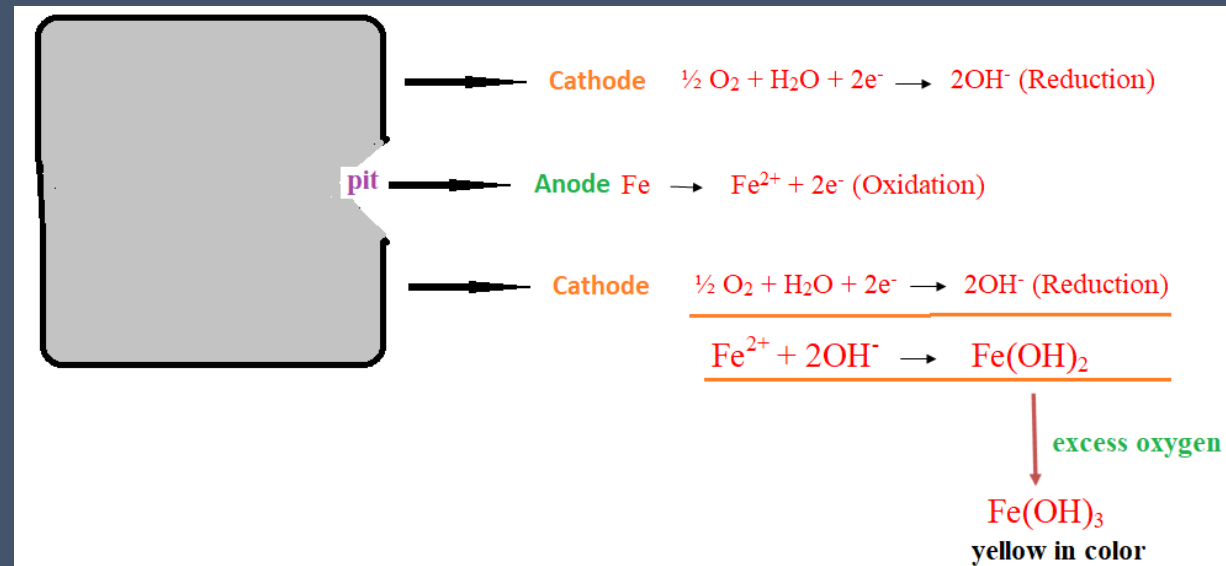
- ❑ occurs between two joining surface (metal-metal or metal-nonmetal crevice)
- ❑ localized corrosion of metal or alloy surface.
- ❑ the joining surface area has low concentration of oxygen as compare to other area.
- ❑ similar to absorption of oxygen step of electrochemical theory of corrosion

Crevice corrosion



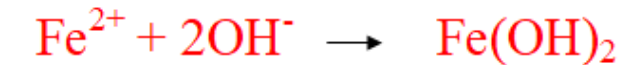
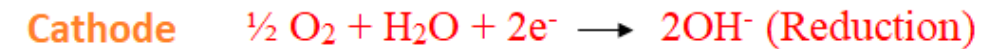
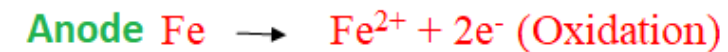
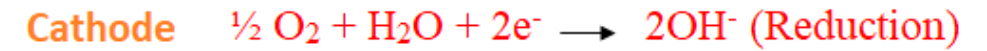
Pitting corrosion

- A pit is formed when the protective coating on the metal surface break, a micro pit (anode) formed on the metal surface.
- Once the pit is form the process of corrosion become very fast due to different amount of oxygen in contact with metal surface.
- The portion (pit) with lower concentration of oxygen become anode and portion with higher concentration of oxygen becomes cathode.

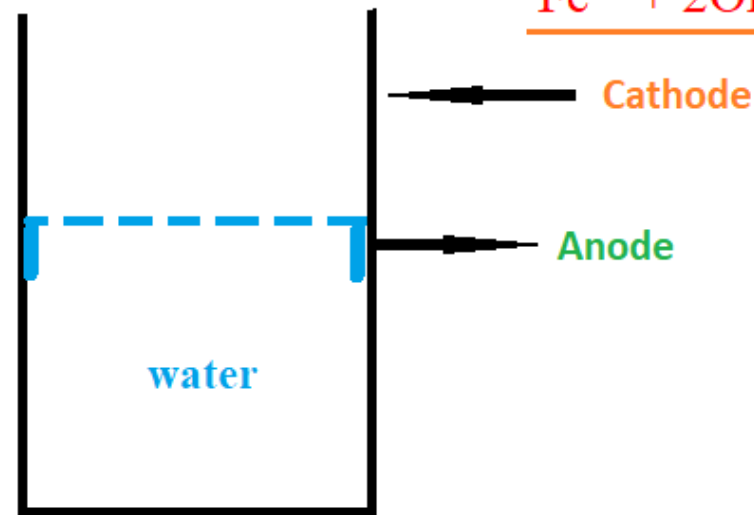
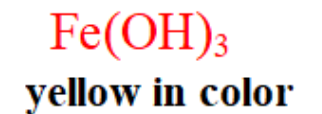


Water-line corrosion

When water is stored in the steel tank, then corrosion starts, along the line beneath the level of water meniscus. As we know that beneath the water level concentration of oxygen is very low therefore metal part below the water level acts as anode and metal part above the water level have higher concentration of oxygen therefore this region act as cathode.



excess oxygen



Microbial Corrosion

Microbial corrosion, also known as microbiologically influenced corrosion (MIC), is a type of corrosion that occurs due to the activity of microorganisms like bacteria, archaea, and fungi. These microorganisms can attach to metal surfaces, alter the local chemical environment, and accelerate the corrosion process.

This form of corrosion can affect various materials, including metals and concrete, and is a significant concern in many industries.



Microbial corrosion or biological corrosion, is the deterioration of metals as a result of the metabolic activity of microorganisms. There are about a dozen of bacteria known to cause microbial corrosion of carbon steels, stainless steels, aluminum alloys and copper alloys in waters and soils with pH 4~9 and temperature 10°C~50°C. These bacteria can be broadly classified as aerobic & anaerobic. Sulphate reducing bacteria (SRB) is anaerobic and is responsible for most instances of accelerated corrosion damages to ships and offshore steel structures. Iron and manganese oxidizing bacteria are aerobic and are frequently associated with accelerated pitting attacks on stainless steels at welds.

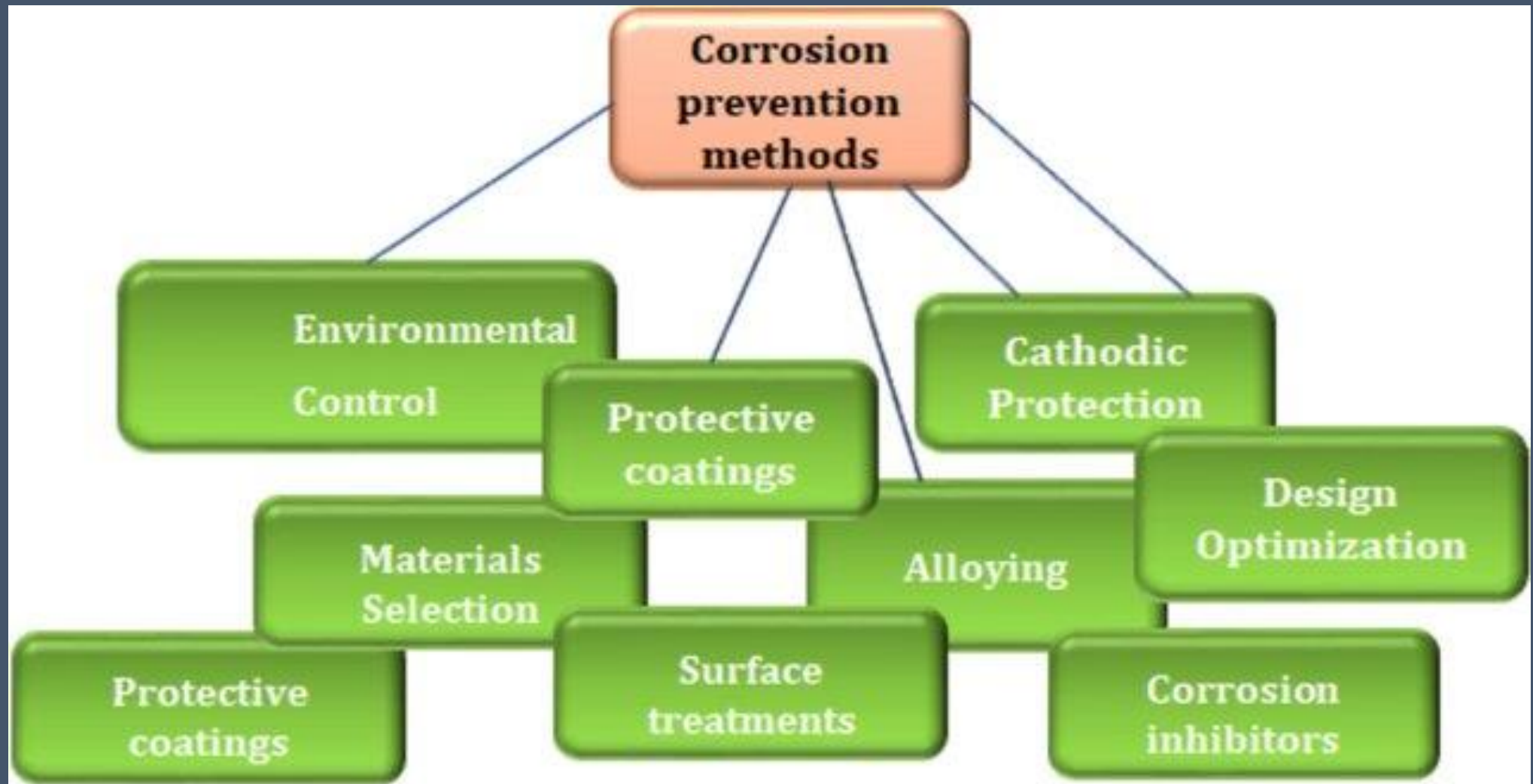
Microbial Corrosion: Anaerobic bacteria

Anaerobic bacteria can utilize various electron acceptors other than oxygen in a process called anaerobic respiration. While some bacteria can perform salt oxidation as part of their metabolism. Salt oxidation is not a common or well-defined term in anaerobic bacterial respiration. Anaerobic respiration in bacteria typically involves the reduction of other inorganic compounds like nitrate, sulfate, or carbon dioxide, or sometimes organic compounds like fumarate.

Factors affecting corrosion

- Reactivity of metal: If metal is more reactive then it undergoes corrosion more readily.
- Strain in metal: Corrosion takes place readily at cuts and bends area of metal.
- Presence of impurities: If impurities present in the pure metal then metal undergoes corrosion more readily.
- Presence of electrolyte: In saline water (electrolyte) metal readily corrodes.
- Air and moisture: Best example rusting of iron.
- pH: $\text{pH} < 7$ is more corrosive than basic or neutral medium.
- Temperature: At higher temperature rate of corrosion become increased.

Corrosion protection



Corrosion protection: Metal Coating

The coating of metal can be achieved by following processes:

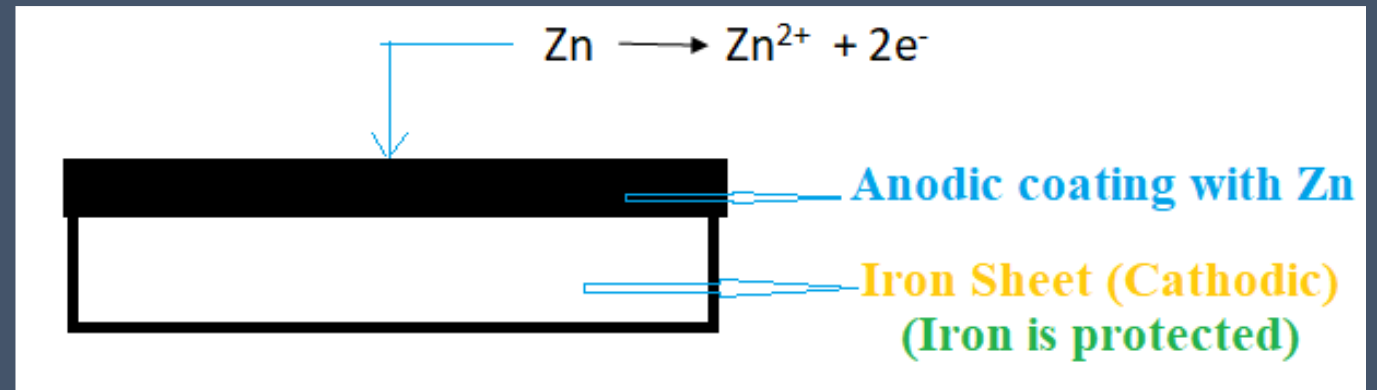
Anodic metal coating (Galvanization)

Cathodic metal coating (Tinning).

Corrosion leading $\text{Fe}(\text{OH})_2$ or Fe_2O_3 form non-compact (porous) layer that allow penetration of O_2 and moisture.

Anodic metal coating
(Galvanization)

Spontaneous reaction



Metal coating which is anodic to base metal, known as anodic metal coating. In order to protect a metal from corrosion, any metal (anodic in nature) which is higher in the galvanic series like Zn is coated on the surface of metal (base metal/cathodic in nature). The best example of anodic metal coating is Galvanization in which coating of base metal (to be protected) takes place by dipping the base metal (cathodic in nature) into the molten zinc (anodic in nature).

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

Half Reaction			Standard Potential (V)
F₂	+ 2e ⁻ ⇌	2F⁻	+2.87
Pb⁴⁺	+ 2e ⁻ ⇌	Pb²⁺	+1.67
Cl₂	+ 2e ⁻ ⇌	2Cl⁻	+1.36
O₂ + 4H⁺	+ 4e ⁻ ⇌	2H₂O	+1.23
Ag⁺	+ 1e ⁻ ⇌	Ag	+0.80
Fe³⁺	+ 1e ⁻ ⇌	Fe²⁺	+0.77
Cu²⁺	+ 2e ⁻ ⇌	Cu	+0.34
2H⁺	+ 2e ⁻ ⇌	H₂	0.00
Pb²⁺	+ 2e ⁻ ⇌	Pb	-0.13
Fe²⁺	+ 2e ⁻ ⇌	Fe	-0.44
Zn²⁺	+ 2e ⁻ ⇌	Zn	-0.76
Al³⁺	+ 3e ⁻ ⇌	Al	-1.66
Mg²⁺	+ 2e ⁻ ⇌	Mg	-2.36
Li⁺	+ 1e ⁻ ⇌	Li	-3.05

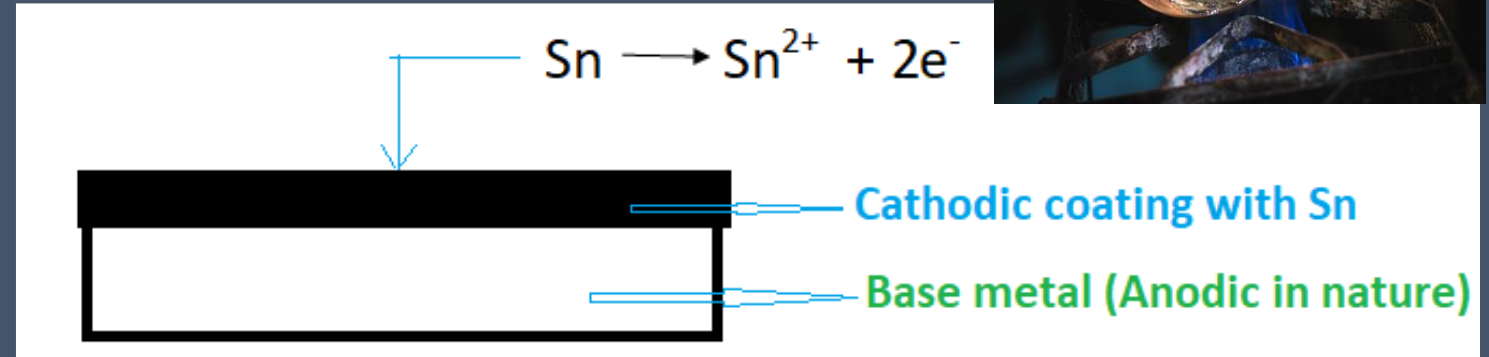
Activity Series of Metals in Aqueous Solution

Metal	Oxidation Reaction
Lithium	$\text{Li(s)} \rightarrow \text{Li}^+(\text{aq}) + \text{e}^-$
Potassium	$\text{K(s)} \rightarrow \text{K}^+(\text{aq}) + \text{e}^-$
Barium	$\text{Ba(s)} \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{e}^-$
Calcium	$\text{Ca(s)} \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{e}^-$
Sodium	$\text{Na(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{e}^-$
Magnesium	$\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$
Aluminum	$\text{Al(s)} \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{e}^-$
Manganese	$\text{Mn(s)} \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{e}^-$
Zinc	$\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$
Chromium	$\text{Cr(s)} \rightarrow \text{Cr}^{3+}(\text{aq}) + 3\text{e}^-$
Iron	$\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$
Cobalt	$\text{Co(s)} \rightarrow \text{Co}^{2+}(\text{aq}) + 2\text{e}^-$
Nickel	$\text{Ni(s)} \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{e}^-$
Tin	$\text{Sn(s)} \rightarrow \text{Sn}^{2+}(\text{aq}) + 2\text{e}^-$
Lead	$\text{Pb(s)} \rightarrow \text{Pb}^{2+}(\text{aq}) + 2\text{e}^-$
Hydrogen	$\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$
Copper	$\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$
Silver	$\text{Ag(s)} \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$
Mercury	$\text{Hg(l)} \rightarrow \text{Hg}^{2+}(\text{aq}) + 2\text{e}^-$
Platinum	$\text{Pt(s)} \rightarrow \text{Pt}^{2+}(\text{aq}) + 2\text{e}^-$
Gold	$\text{Au(s)} \rightarrow \text{Au}^{3+}(\text{aq}) + 3\text{e}^-$

Corrosion protection: Metal Coating

Cathodic metal coating (Tinning)

Spontaneous reaction



Metal coating which is cathodic to base metal, known as cathodic metal coating.

In order to protect a metal from corrosion, any metal (cathodic in nature) which is lower in the galvanic series like Sn, Cu, Ni etc., are coated on the surface of metal (base metal/ anodic in nature).

The best example of cathodic metal coating is Tinning. Tinning is a process in which coating of base metal (to be protected) takes place by dipping the base metal (anodic in nature) into the molten Tin (Sn) (cathodic in nature).

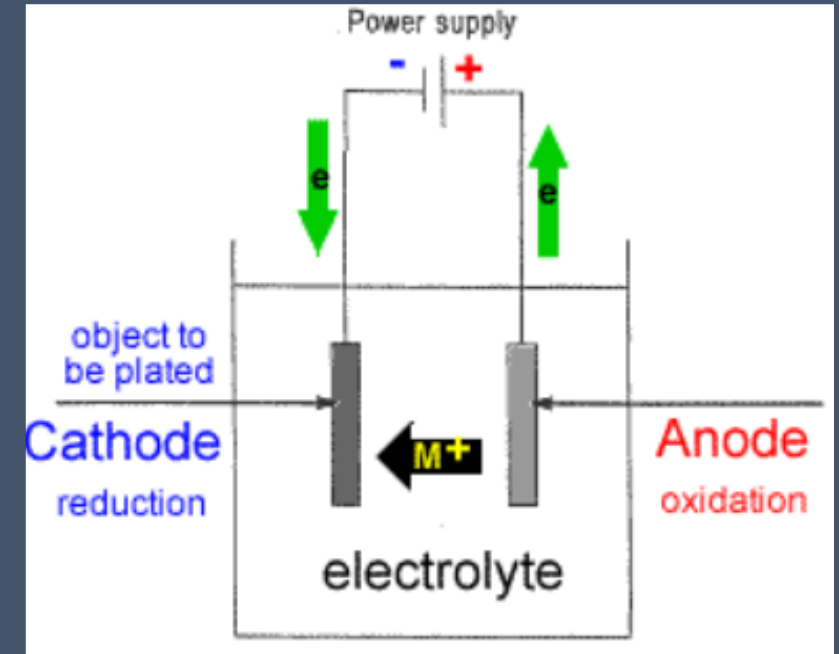
Corrosion protection: Electroplating

Electroplating

In electroplating, electro-deposition of metal (anode) by means electrolysis over surface of metals and alloys (cathode or base metal). The two electrodes are dipped in the electrolyte solution which contains the metal ions to be deposited on the base metal.

When a direct current is passed from an external source, the coating metal ions migrate towards cathode and get deposited over the surface of base metal in the form of a thin layer. In electroplating metal like Au, Ag, Zn, Sn, Cr, Ni etc. can be placed on the surface of base metal (Cathode).

Non-spontaneous reaction



Corrosion protection

Corrosion leading Fe(OH)_2 or $_{/3}$ form nonc-compact (porous) layer that allow penetration of O_2 and moisture. On the other hand, protecting metal (coated metal) is generally form compact oxide layers on its own surface that does not allow penetration of reacting chemicals like O_2 , moisture etc. The protecting metals are generally Zn, Al, Sn, Au etc.